

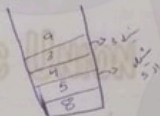
Q1) Multiple Choice (5 pts)

Choose the correct answer in the following

1. after execution the following code, the context of stack is

```
stack<char> s;  
s.push(8);  
s.push(5);  
s.pop();  
s.push(4);  
s.push(3);  
s.pop();  
s.push(9);
```

LIFO



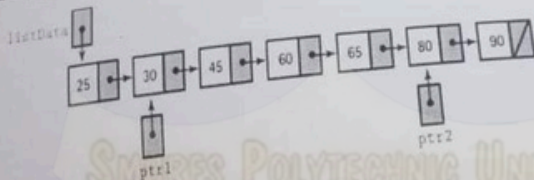
- a. 8,5,4,3,9
b. 8,4,3,9
c. 3,9
d. 8,4,9

2. what is the output of the following code:

```
template <typename T, typename k >  
K add(T num1, k num2) {  
    return (num1 + num2);  
}  
int main() {  
    cout<<add<int,float>(2.5, 3.7);  
}
```

- a. 5.5
b. 5.7
c. 5
d. 6.2

Q2: Consider the following linked list then answer the below questions: (8 pts)



A. Are the following expressions true or false

1. listData->next == ptr1 (✓) 2
2. ptr1->next->info == 60 (✓) 2
3. ptr2->next == NULL (✓) 2
4. listData->info == 25 (✓) 2

B. Write one statement to do each of the following:

1. Make listData point to the node containing 45.

~~ptr1 -> next = 45~~

2. Make ptr2 point to the last node in the list.

~~ptr2 -> next = 90~~

3. Make listData point to an empty list.

~~ListData = null~~

4. Set the info member of the node 45 to 60.

~~ptr2 -> next -> info = 60~~

Q3) a-fill in the blanks (7 pts)

```

#include <iostream>
using namespace std;

class Test {
public:
    float r, area;
    void input();
    // prompt the user to enter the radius
    // enter the radius
    void findArea();
    // calculate the area
    void display();
    cout << "Area of circle is:" << area;
};
  
```

```

int main() {
    // public obj
    obj.....
    obj.....
    obj.....
    obj.....
    return 0;
}
  
```

3.

after execution the following code, the rear and front values are ---

```
queue<char> q;
q.enqueue(5);
q.enqueue(4);
q.dequeue();
q.enqueue(9);
q.dequeue();
```

- a. rear=2 front=2
- b. rear=3 front=2
- ☒ c. rear=1 front=2
- d. rear=2 front=1

4. trace the following link list code:

```
void fun(node* head, int x)
{
    node* current = head;
    int c = 0;
    while (current != NULL) {
        if (current->data != x)
            c++;
        current = current->next;
    }
    cout << c;
```

Assume: X=5

- a. 2
- b. 5
- c. 3
- ☒ d. 4

5. if stack S = 8,5,2,1 what is the content of the stack S after executing the following function

```
template <class T>
void stack<T>::fun(){
    for(int i=tos; i>=0; i--){
        if (array[i] > 4)
            array[i]++;
    }
}
```

- ☒ a. 9 6
- ☒ b. 9 6 2 1
- c. 3 2
- d. 8 5 3 2

8, 5, 2, 1